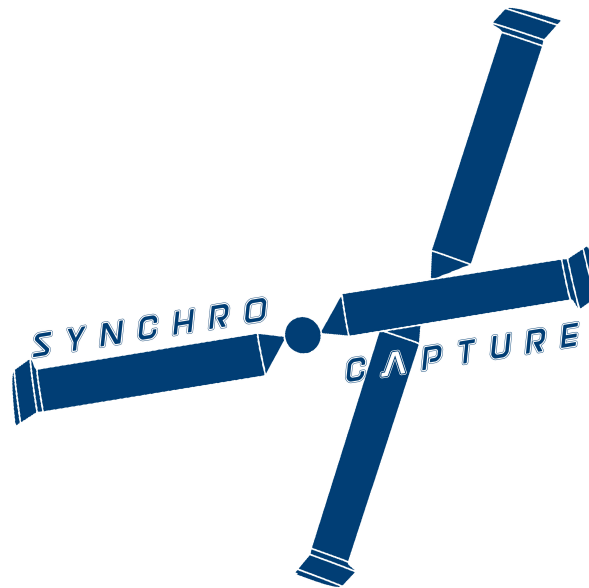


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Engine Design For A Novel Synchropter For The Aerial Capture Of An Orbital Launch System



Student: Omar Arangath-Joseph
CID: 01347612

Project Supervisor: Dr. E. Levis
Team Supervisor: Dr. M. Ribera
Submission Date: 15/06/2020

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ

Abstract

With the increased importance of commercial spaceflight, methods of reducing costs of orbital launches are being developed, and ensuring full reusability of the launch vehicle is an effective way of doing this. Reusability can be achieved by the use of a recovery vehicle being able to aerially capture the rocket stage before impact with the ground. The scope of this report is to focus on the work carried out in designing the recovery vehicle's powerplant.

The main design choices and compromises made are covered in the Conceptual Design phase, after which the Preliminary Design phase deals with the design of various components of the powerplant, as well as briefly evaluating the performance and assessing the ability of the powerplant to meet the requirements that were set out.

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath - Joseph Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

Contents

I	Conceptual Design Of A Helicopter Powerplant	1
1	Pre-Conceptual Phase & Conceptual Design Introduction	1
2	Rotor sizing & Power Requirements Estimation	1
2.1	Baseline Configuration	1
2.2	Mission Profile	1
2.3	Initial Power Requirement Estimate & Rotor Sizing	1
2.4	Powerplant Comparison Selection	3
2.5	Decision On Powerplant Type	3
2.5.1	Non-Diesel Piston	3
2.5.2	Electric	3
2.5.3	Diesel & Turboshaft	3
2.5.4	Consideration of Hybrid System	4
2.6	Turboshaft Cycle Design	4
2.6.1	Overall Engine Configuration	4
2.6.2	Initial Parametric Cycle Study	5
2.6.3	Cycle Calculation	6
II	Preliminary Design	7
3	Preliminary Design Introduction	7
4	Refining the Cycle Design	7
4.1	Turbine Cooling Requirements	7
5	High Pressure Turbine Design	8
5.1	Introduction and Design Goals	8
5.2	Design Process	8
5.3	Design Results	9
6	Compressor Design	10
6.1	Introduction	10
6.2	Design Process	10
6.3	Design Results	10
7	Power Turbine Design	11
7.1	Design Process Overview	11
7.2	PT Design Results	11
8	Discussion, Conclusions & Overall Dimensions	12
	References	14

A	Pre-Conceptual Design	17
A.1	Pre-Conceptual Design Introduction	17
A.2	Baseline Configuration	17
A.3	Initial Sizing Calculations	17
B	Power Estimation Calculations	17
B.1	Hovering Flight	18
B.2	Vertical Climb	18
B.3	Forwards flight	18
C	Conceptual Engine Design	19
C.1	Engine Type Comparison	19
C.2	Electric Engine Weight Estimate	19
C.3	Comparison of Piston and Turboshift Engines	20
C.4	Hybrid System Calculation	20
C.5	Turboshift Configurations	21
C.5.1	Single Spool	21
C.5.2	Dual Spool	21
C.5.3	Three Spool	21
D	Initial Cycle Design	21
D.1	Initial Cycle Design Script	21
D.2	Turbine Cooling Details	23
E	Turbine Design & Analysis	24
E.1	Turbine Graphs	24
E.2	HPT Design Results & Dimensions	25
F	Compressor Design & Analysis	27
F.1	Compressor Graphs	27
F.2	Compressor Design Results & Dimensions	29
G	Power Turbine Results & Dimensions	30
H	GasTurb 13 Diagrams	32
H.1	Performance	32
H.2	Other	32
H.3	Compressor and Turbine Maps	33

List of Tables

1	<i>Team Members and Roles.</i>	ii
2	<i>Weight estimates of diesel and turbine propulsion systems</i>	4
3	<i>Results from initial cycle design</i>	6
4	<i>Turbine cooling mass flow requirements</i>	8
5	<i>Cycle parameters from GasTurb13</i>	8
6	<i>Key results of HPT design</i>	9
7	<i>Key results of Compressor design</i>	11
8	<i>Key results of PT design</i>	12
9	<i>Summary of Key Vehicle Specifications</i>	13
10	<i>Comparison of rotorcraft powerplant types</i>	19
11	<i>Power to weight ratios of helicopter piston (P) and turbine (T) engines.</i>	20
12	<i>Final HPT Design Results</i>	26
13	<i>Final HPT Design Dimensions</i>	26
14	<i>Final Compressor Design Parameters</i>	29
15	<i>Final Compressor Design Dimensions</i>	29
16	<i>Final PT Design Results</i>	30
17	<i>Final PT Design Dimensions</i>	30

List of Figures

1	<i>Effects of varying disc loading on power requirements and rotor diameter.</i>	2
2	<i>Effect of varying disc loading on power requirement assuming no power lapse.</i>	2
3	<i>Overall weight of propulsion system as a function of electric boost motor power.</i>	4
4	<i>3 main types of turboshaft designs[1]</i>	5
5	<i>Specific fuel consumption and specific power as a function of burner exit temperature and compressor pressure ratio, with contours showing engine core efficiency</i>	6
6	<i>HPT temperature ratio and α_3 as a function of α_2.</i>	9
7	<i>Attainable temperature ratios and α_3 values as a function of α_2 and Ω.</i>	12
8	<i>Effects of varying inlet mach number relative to blades M_2 on outlet swirl angle α_3 and rotor reaction degree R_r</i>	24
9	<i>Effects of varying inlet mach number relative to blades M_2 on stage temperature ratio</i>	24
10	<i>Effects of varying Ω on various turbine parameters</i>	25
11	<i>Effects of varying Ω on various turbine parameters</i>	25
12	<i>Diagram explaining different turbine dimensions.[2, p 290]</i>	27
13	<i>Effects of varying D on various compressor parameters</i>	27
14	<i>Effects of varying M_1 on pressure ratio</i>	28
15	<i>Effects of varying σ on various compressor parameters</i>	28
16	<i>Engine performance (SFC & power available) as a function of altitude and temperature.</i>	32
17	<i>Power variation with RPM and fuel flow.</i>	32
18	<i>Key Engine Dimensions</i>	33
19	<i>Compressor map</i>	33
20	<i>High Pressure Turbine Map</i>	34
21	<i>Power Turbine Map</i>	34

Part I

Conceptual Design Of A Helicopter Powerplant

1 Pre-Conceptual Phase & Conceptual Design Introduction

Before a rotorcraft design was selected, the first phase of the project involved three separate sub-groups assessing the feasibility of different recovery vehicles. As the work carried out in this phase was focused on a fixed-wing design, which was not selected in the end, this has been omitted from the report but is available in Appendix A. This section will cover the research and conceptual design choices made during the rotorcraft powerplant design. Before a powerplant type could be selected, the baseline configuration and mission profiles had to be determined, and an estimate of power requirement was also needed. Several powerplant options were considered to begin with and compared in order to make a choice, with a turboshaft engine being selected in the end. Note that this report focuses on the design of the rotating turbomachinery, for details on burner design see [3], for inlet, exhaust and lubrication system design see [4], and for structural integration see [5] and [6].

2 Rotor sizing & Power Requirements Estimation

2.1 Baseline Configuration

The baseline configuration of the rotorcraft was selected to be a synchropter design (see [7] & [8]), the main implication of this on powerplant design being that no power for the tail rotor is required. It was also decided that the design was to be unmanned, which reduced the amount of regulations needed that needed to be complied with [9] and made single engine operations desirable in order to fit the propulsion system inside the small fuselage to maintain good aerodynamic performance.

2.2 Mission Profile

Another important aspect required in order to make a decision on the powerplant type was the mission profile and the performance requirements it entails. The mission profile requires the helicopter to climb up to 20,000ft [8], which has an important impact on the powerplant due to the significant power lapse encountered at these altitudes for certain engines. Another requirement was an endurance of 2 hours in order to fly to the rocket's landing site (which could be up to 50 km from the ship) and back.

2.3 Initial Power Requirement Estimate & Rotor Sizing

In order to have an understanding of the power output required throughout the mission, the methods described in Prouty's Helicopter Performance [10] were used (see Appendix B) to estimate power requirements. As data from the engine and MTOW must be iterated together until converged, at this stage the MTOW was obtained from regression data by other members of the systems team ([3] & [4]), with the result being 3600kg. Once a powerplant type was selected the data could then be used to converge the MTOW.

The power requirement of a helicopter is also highly dependent on the disc loading, which is the thrust provided by the rotor divided by its total swept area. As the disc loading had important

implications on the design as a whole, it was decided to choose a target value in collaboration with the Aerodynamics team ([11] & [12]). The impact of disc loading on the required rotor diameter and power was investigated using the methods from Prouty[10]. The results are displayed in figure 1, where the power requirements refer to hover at MTOW at 20,000ft (this requirement was later deemed to be excessive and was revisited, as explained in section 2.5.4).

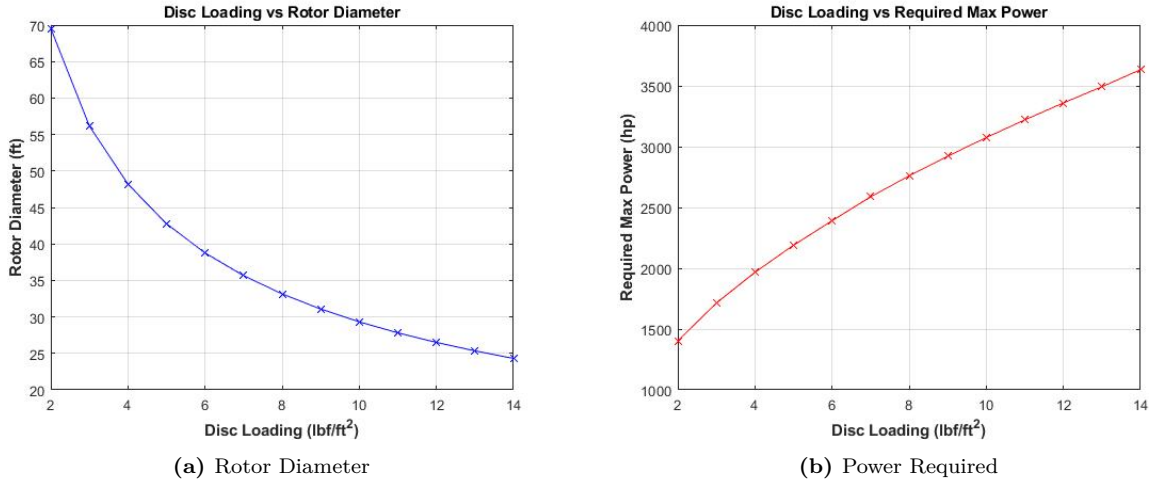


Figure 1: Effects of varying disc loading on power requirements and rotor diameter.

The above power requirements assume an altitude power lapse consistent with turboshaft engines. For an engine that does not experience power lapse, the power requirement is shown in figure 2 below.

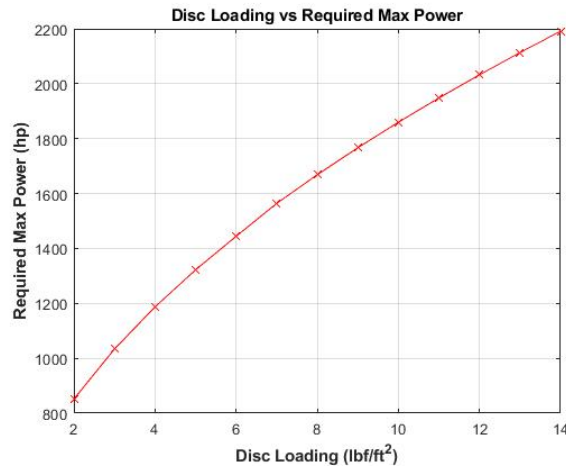


Figure 2: Effect of varying disc loading on power requirement assuming no power lapse.

In collaboration with the Aerodynamics team ([11] & [12]), it was decided to select a disc loading of 6lb/ft², as this provided a good compromise between power requirement and rotor size. Thus the power requirement is 2400 HP if power loss is present, or 1450 HP in case of no power lapse.

2.4 Powerplant Comparison Selection

For a rotorcraft, there are 3 main types of engine available: piston, turboshaft, or electric. A comparison of the advantages and disadvantages of each is presented in table 10 in Appendix C.1. In general, turboshaft designs have the highest power to weight ratio and are more reliable, but are more expensive and fuel consuming than piston engines. The electric design has the advantage of suffering no power lapse with altitude, but the low energy density of batteries available leads to a significant range or weight penalty. In addition the possibility of a diesel piston engine was considered. The advantages and disadvantages are the same as for the standard piston engine, with the exception of power lapse, as a diesel engine suffers no significant power lapse up to very high altitudes (30,000ft[13]), thus making them an attractive choice in this case.

2.5 Decision On Powerplant Type

2.5.1 Non-Diesel Piston

This option was quickly discarded as it offered no real benefits compared to a diesel piston, and the high power lapse (greater than turboshafts) makes it ill-suited for high altitude operations without modifications such as superchargers. These however add additional weight and complexity, which is undesirable given the already low power to weight ratio and reliability in comparison to turboshafts, especially as the design is to operate on a single engine.

2.5.2 Electric

While an electric design is certainly desirable due to suffering no power lapse and in terms of innovativeness, the still large power requirement in combination with the 2 hours of endurance required would likely result in excessive battery mass, despite the relatively low weight of the motor itself. To investigate this, a calculation was carried out using an average estimated power of 360kW (obtained from taking the average power requirement for all mission segments) throughout the 2 hours and taking the highest energy density of commercially available Lithium-ion batteries (approximately 1MJ/kg[14]) results in an estimated battery weight of over 2.5 tonnes (see Appendix C.2 for details), which is unacceptably high as it is over two thirds the weight of the helicopter. Hydrogen fuel cells could serve as an alternative, but their power to weight ratio, although far better than batteries, is still far lower than turboshafts[15] and along with several other complications[16] their use in heavy rotorcraft would be prohibitive. This is in addition to all the complications of using hydrogen as a fuel especially if considering operations from a ship. As a result, a fully electric design was scrapped.

2.5.3 Diesel & Turboshaft

The primary advantage of the diesel is the lack of power lapse and lower fuel consumption, however the turboshaft provides a much higher power to weight ratio and higher reliability. The power to weight ratios for several turboshaft and piston engines are compared in table 11 in Appendix C.3. A calculation to estimate the weight of each of these two propulsion systems was carried out, taking into account fuel and engine weight, (see Appendix C.3 for details) the results of which are shown in table 2 below.

Table 2: *Weight estimates of diesel and turbine propulsion systems*

	Diesel	Turboshaft
Estimated Engine Weight (kg)	833	514
Estimated Fuel Weight (kg)	159	406
Total Weight (kg)	992	920

As can be seen, despite the lower power requirement the diesel engine is still significantly heavier than a turboshaft due to the much lower specific power (taken as 1.2kW/kg for the diesel and 3.5kW/kg for the turboshaft, which is the worst case for turbine and best case for diesel, see Appendix C.3). Although the diesel uses less fuel (SFC 0.22kg/kWh for diesel and 0.34kg/kWh for turbine, from Appendix C.3), overall the turboshaft configuration is slightly lighter. In addition, the less compact size that a diesel engine of such size would have caused problems with the integration into the small streamlined fuselage that resulted from the choice to make the helicopter unmanned. This together with the lower weight and better reliability resulted in the turboshaft being chosen.

2.5.4 Consideration of Hybrid System

Once the turboshaft had been chosen, the MTOW of the design was converged to 2907kg([3],[4]). During this process, it was decided to size the engine for hovering with the payload at 15,000 instead of 20,000ft, as the previous case lead to high excess power that was seldom used. This in addition to the lower MTOW lead to a lower power requirement of 1600 HP instead of the original 2400. This change has not affected the decisions taken so far however, as although overall power requirement has lowered due to lower MTOW, the power requirement of the turboshaft has decreased far more than that for diesel and electric engines due to the large reduction in power lapse. As a result the turboshaft would have a larger weight advantage over the diesel, while an electric design would still require almost 2 tonnes of batteries, using the same calculation as before.

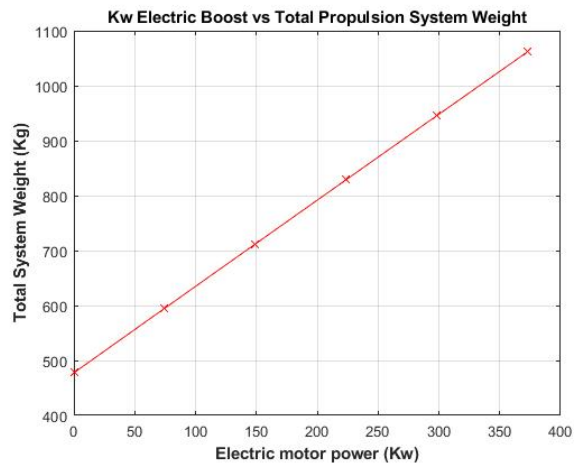
The possibility of using an electric motor to supplement the power lost at high altitude was then investigated. The advantage of this is to lower the power requirement of the main engine thus decreasing the required fuel and engine weight.

However, battery mass and a more complex transmission design in order to accommodate both systems would also be required. A simple trade off study was done (see Appendix C.4 for details), the results of which are displayed in figure 3. These show that the additional battery mass increases the overall weight despite the reduced engine and fuel weight. As a result of the lack of potential benefits and increased complexity, the hybrid design was also dropped.

2.6 Turboshaft Cycle Design

2.6.1 Overall Engine Configuration

Now that the engine type had been decided, the first decision to be made was selecting the configuration of the turboshaft engine. There are 3 main options, presented in figure 4 below.

**Figure 3:** *Overall weight of propulsion system as a function of electric boost motor power.*

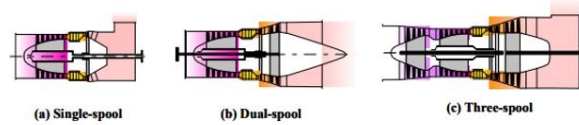


Figure 4: 3 main types of turboshaft designs[1]

A full breakdown of the differences and advantages & disadvantages of each design is given in Appendix C.5, but in general as the complexity increases so do the cost but also the performance and flexibility achievable. At this stage the decision was taken to evaluate the feasibility of using a two spool design, as this provides a good compromise between cost and performance, however if an acceptable performance could not be achieved, then a three spool design would be considered. Another decision to make was whether to use a fully axial design or a combination of axial and radial rotors as is often the case in turboshaft designs. The primary advantages of a radial design are the higher compressor pressure ratios achievable per stage, meaning less stages and thus less weight is required. However, they also result in a greater engine cross-sectional size and the difficulty of "stacking" stages together means that they must be used together with axial stages. Additionally, at flow rates larger than about 1.4 kg/s (which is certainly the case on an engine of this size) they suffer from lower efficiency [1, p. 34]. Also, due to the rather unusual requirement for a helicopter to fit the engine inside the fuselage for lower drag, instead of mounting it in a fairing on top of the fuselage, it was desirable to minimise cross-sectional area, so a fully axial design was selected.

2.6.2 Initial Parametric Cycle Study

The first step of the cycle design was to decide on values for the compressor pressure ratio and the burner exit temperature, which are the main factors in determining engine performance. Using the GasTurb13 software, figure 5 (shown next page) was produced. Using this graph, it was found that the two spool configuration chosen could provide the required power at reasonable SFCs ($< 0.36\text{kg/kWh}$) and mass flow rates ($< 6\text{kg/s}$) (similar values to existing engines[17]) even with moderate pressure ratios and burner temperatures, hence there was no need for the extra complexity of a three spool design.

However in the interests of creating a modern and competitive design, it was decided to take advantage of developments in high temperature alloys and blade cooling technology. This allows the burner exit temperature to be raised to higher values, and as can be seen from figure 5, this has a very beneficial impact on engine performance. Although manufacturing costs for this are higher, it was reasoned that the significantly improved fuel efficiency would over the lifetime of the engine lead to lower operating costs, and as any temperature over 1400K would require this technology in any case, it was decided to try and maximise the burner exit temperature. A value of 1700K was thus chosen, as any higher than this the limits of even the most advanced alloys are exceeded without significant amounts of cooling, greatly increasing manufacturing costs. As for the compressor pressure ratio, as can be seen by figure 5 again higher values result in better performance, however above 10 there are diminishing returns for this configuration. As a result a value of 10 was selected as a good trade-off in terms of the complexity and cost of the compressor required for the increase in performance.

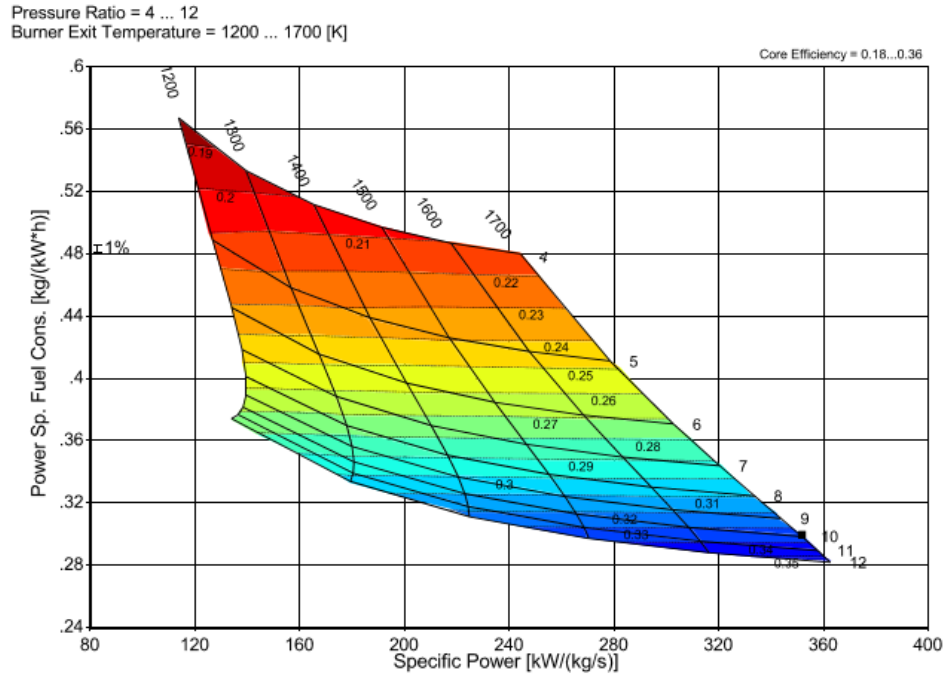


Figure 5: Specific fuel consumption and specific power as a function of burner exit temperature and compressor pressure ratio, with contours showing engine core efficiency

2.6.3 Cycle Calculation

Now that the layout and key engine parameters had been selected, the overall cycle could be calculated. At this stage this was done using a MATLAB script which is available in Appendix D.1, using basic thermodynamics with real effects accounted for by using the adiabatic efficiencies of components, following the methodology used in [1]. In order to keep the calculation results more conservative, the efficiencies were assumed at the lower end of normal values[1, p 27-45]. The resulting temperature and pressure ratios required to obtain an output power of 1250kW (1200kW power requirement plus an additional 50kW for electrical generation) are shown in table 3.

Table 3: Results from initial cycle design

Parameters	Inlet	Compressor	Burner	HPT	PT	Exhaust
Total temperature ratio	0.90	2.09	3.00	0.84	0.80	0.96
Total pressure ratio	0.97	10	0.92	0.37	0.31	0.98
Adiabatic efficiencies	N/A	0.85	N/A	0.85	0.85	N/A

Note that while the inlet and exhaust adiabatic efficiencies can also be defined, they are not required in this calculation and have hence been omitted. The resulting mass flow rate of air is 3.57 kg/s, with 0.12 kg/s of fuel for a fuel to air ratio of 0.032 and a power specific fuel consumption of 0.35 kg/kWh.

Part II

Preliminary Design

3 Preliminary Design Introduction

Now that the engine cycle has been defined, a higher fidelity model of the engine cycle using GasTurb 13 can be obtained, followed by the preliminary design of the various components.

4 Refining the Cycle Design

4.1 Turbine Cooling Requirements

Before using GasTurb 13 to analyse the cycle, the amount of compressor bleed air required to cool the turbines needs to be estimated, as this is used as an input. In order to do this the methodology from [1, p 167-171] was used, and is outlined here. First, the maximum allowable metal temperature for guide vanes and rotors was estimated, using the following empirical relations:

$$T_{m(vane)} = [10(year) - 17640] - [100\log(life) - 400] \quad (1)$$

$$T_{m(blade)} = [10(year) - 17740] - [100\log(life) - 400] \quad (2)$$

Where *year* represents the year the material has entered service, and *life* is the desired lifetime of the component in hours. Note that the output of the equations is in degrees Rankine. In order to reduce material costs, it was decided to compromise between using the newest but costlier materials and the required cooling mass flow. Thus, the material year selected was 2000, as for a design lifetime of 20,000 hours this yields a maximum allowable temperature of 1300K for the guide vanes and 1240K for the blades. If this were to require excessive cooling, a newer material could be chosen and the cooling estimate calculations redone.

The next step is to calculate the cooling effectiveness factor E .

$$E = \frac{T_{entry} - T_m}{T_{entry} - T_{cooling}} \quad (3)$$

Where T_{entry} is the temperature at the turbine inlet, taken to be equal to the burner outlet temperature with a safety factor of 100K added, T_m is the maximum allowable metal temperature, calculated from equations (1) and (2), and $T_{cooling}$ is the cooling flow temperature, assumed to be equal to the compressor outlet temperature. The ratio of cooling mass flow required to total mass flow is then obtained using equation (4).

$$\frac{\dot{m}_c}{\dot{m}} = 0.222F \left(\frac{E_{PF}}{1 - E_{PF}} \right) \quad (4)$$

Where E_{PF} is E corrected for the pattern factor as explained in Appendix D.2, and the cooling factor F depends on the cooling configuration is used. A value of 1 was used corresponding to the widely used blade film cooling configuration[1, page 171], in which the cooling air is passed through small holes throughout the blade creating a film of cooler air surrounding the blade. If this were to be found insufficient, a more advanced cooling method could be selected. The results of this equation are increased by a factor of $\frac{4}{3}$ to account for various factors such as wall & disc cooling and leakage. The results obtained are shown in table below.

Table 4: *Turbine cooling mass flow requirements*

	HPT Stator	HPT Rotor	PT Stator	PT Rotor	Total
Cooling mass flow ratio (%)	3.69	3.62	0.73	0.36	8.40

The final result from this calculation was that a total of 8.4% of the compressor mass flow was needed for cooling, which is equivalent to 0.3 kg/s. This was deemed an acceptable result as it falls in the mid-range of standard turbine cooling requirements[18]. The data was thus used in the GasTurb 13 software to generate the new cycle data. The final results are similar to those obtained from the initial cycle analysis, and are shown in table 5 below.

Table 5: *Cycle parameters from GasTurb13*

Parameters	Inlet	Compressor	Burner	HPT	PT
Total temperature ratio	0.90	2.07	2.84	0.80	0.79
Total pressure ratio	0.97	10	0.92	0.4	0.30
Polytropic efficiencies	N/A	0.89	N/A	0.84	0.83

The adiabatic efficiencies are the same as before and used as inputs hence not shown, while the inlet temperature/pressure ratios, compressor pressure ratios, and burner exit temperature are also inputs and hence unchanged. The finalised cycle has an air mass flow rate of 3.46 kg/s, giving a required total cooling flow rate of 0.29 kg/s and a fuel flow rate of 0.10 kg/s for a fuel to air ratio of 0.032 (pre-cooling flow) and 0.029 (post-cooling) and an SFC of 0.30 kg/kWh. GasTurb13 also gives more accurate values of γ and C_p for each section, which were then used in the component design. GasTurb was also used to provide performance graphs under various conditions, which were used by the Flight Mechanics[19] and Aerodynamics[12] team to calculate various performance metrics. The engine performance graphs are available in Appendix H.

5 High Pressure Turbine Design

5.1 Introduction and Design Goals

Due to the elevated temperatures that the HPT operates at, its design represents the limiting case of the high pressure spool in this case. As a result the design RPM of the high pressure spool was based on the HPT, which thus had to be designed first. Due to the temperature, single-crystal nickel superalloys are the only viable metals available in this case, and due to their high cost and high density it is desirable to keep the number of stages to a minimum.

5.2 Design Process

The first step was to analyse the requirements of the HPT to determine the number of stages. Using the methods presented in Mattingly et al.[2, page 270-297], the effect of varying the flow inlet angle (α_2), the inlet and outlet Mach numbers relative to the blades (M_2 and M_{3R}), and the non dimensional rotational speed (Ω) on the outlet swirl angle (α_3), stage temperature ratio (τ_s), and rotor degree of reaction (R_r) were obtained and are plotted in the graphs in Appendix E.1.

Increasing the inlet Mach number has a beneficial effect in both reducing α_3 (which must remain under 40° under accepted design convention) and τ_s (note that as $\tau_s < 1$ a lower value corresponds to a greater temperature drop) for a given α_2 . M_2 must be supersonic for the first stage of a turbine, however the upper bound is limited by choking of the guide vanes. M_{3R} must be subsonic however

higher values are also beneficial and thus the highest generally feasible value of 0.9[2, p 273] has been used. Ω , which is calculated as shown in equation (5), is instead limited by material performance.

$$\Omega = \frac{\omega r_m}{\sqrt{g C_p T_o}} \quad (5)$$

Where ω is the the rotational velocity in rad/s, r_m is the turbine mean radius, g is the gravitational acceleration, and T_o is the total temperature at the turbine inlet. Based on the maximum allowable metal temperature evaluated in equations (1) and (2), a list of alloys was consulted[20], and PWA 1484 Single Crystal Nickel-based Superalloy was selected, as it can withstand a temperature of up to 1366K with a strength at this temperature of 311 MPa. Using this data, the maximum allowable mean rim speed is given by (6).

$$\omega r_{max} = \sqrt{\frac{4\sigma_d}{\rho}} \quad (6)$$

Where σ_d is the material yield strength at the operating temperature in lbf/ft², and ρ is the material density in slug/ft³. In this case this yields a maximum allowable mean rim speed of 1222 ft/s (372.5 m/s), and thus $\Omega = 0.292$. Using these parameters the graph shown in figure 6 is plotted.

Z is the Zweifel Coefficient, which is explained in Appendix E.1, however a standard value of 1[1, p 116] has been assumed. As can be seen, the required temperature ratio of 0.80(35) can be achieved at $\alpha_2 \approx 75.5$ and $M_2 = 1.2$, for which $\alpha_3 \approx 36^\circ$, which although quite high falls within the generally accepted range. It was thus decided to use a single stage for the HPT, reducing the weight and cost of the engine. A high α_2 also has the benefit of reducing solidity, meaning less blades are required, reducing cost.

5.3 Design Results

The previously determined initial parameters were then used in the design methods described in Mattingly et al.[2, page 270-297] and refined to exactly match the required temperature ratio, with a more detailed analysis method from [21] also used which was essential in ensuring that the flow at the stator was not choking, due to the relatively high choice of M_2 . A full table of the results and turbine design dimensions obtained are available in Appendix E.2, while the key results are shown here in table 6. The estimated weight and length of the HPT are 48.6kg and 0.087m.

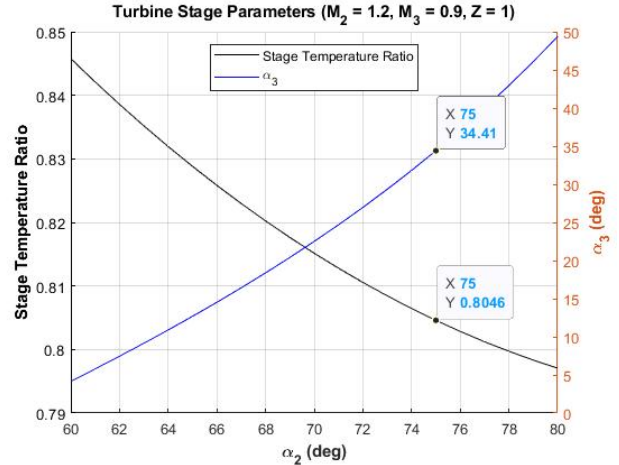


Figure 6: HPT temperature ratio and α_3 as a function of α_2 .

Table 6: Key results of HPT design

Inputs		Outputs	
M_2	1.2	α_3	37.6
M_{3R}	0.9	Mean radius r_m	0.1393
α_2	76.0	RPM	28,100
ωr_{max} (m/s)	372.5	ωr_r actual (m/s)	350.3

The final step is to validate the design by ensuring that the mean wheel rim speed is under the maximum value, which was achieved as shown in table 6. In addition, the Disc Shape Factor (DSF) was calculated using equation (7).

$$DSF = \frac{\rho(\omega r_r)^2}{2\sigma_d} \quad (7)$$

The DSF essentially determines the required maximum rim width in order to support the structural loads. In practice it is to be kept under 2, as the required rim width increases exponentially with DSF. The obtained value from the HPT calculation is 1.77. An additional analysis of material stress is carried out through the AN_{max}^2 parameter, which was found to be less constraining and is described in equation (21) in Appendix E.2.

6 Compressor Design

6.1 Introduction

Having determined the high pressure spool RPM, the design process explained in [2, p 253-268] was used, using the repeating stage model in which the degree of reaction of rotor and stator is split equally. In general due to the much lower temperatures at the compressor the material selection presents much less of a challenge, hence the design could be completed with aluminium 2618-T6 alloy for the first 5 stages, and UNS 56620 Titanium alloy for the last 2 stages.

6.2 Design Process

As in the turbine design process, several parameters were plotted in order to analyse their impact on the performance. In particular the effects of varying the Diffusion factor D , the solidity of the rotors σ , the inlet flow angle α_1 and the inlet Mach number M_1 were analysed. The resulting graphs are shown in Appendix F.1. In general increasing these parameters is beneficial in terms of the performance, however limits are of course imposed by various issues.

In the case of M_1 , it must be low enough to avoid supersonic flow at the compressor tip, as this causes significant losses. The Diffusion factor is used as a descriptor of the technology level of the compressor, with the most advanced axial compressors reaching values of 0.55-0.6[2, p 256]. For the design, a Diffusion factor of 0.5 was used, as it was assumed that as a new company it was unlikely the design would be at the leading edge of the industry, especially given the highly specialised CFD used in modern turbomachinery designs the details of which are highly valuable intellectual property of the major aero engine corporations. As for the rotor solidity, it is usually in the range of 1-1.5, as too low values result in excessive airfoil angle of attack leading to stall, while too high values tend to increase friction losses. As a result the resulting data from using various combinations of M_1 , α_1 and σ was analysed to find an optimum combination that can achieve the required performance whilst maintaining reasonable blade pitch and tip Mach numbers.

6.3 Design Results

In order to determine the required number of stages, the method used is to calculate the total temperature rise required across the compressor, and then use the accepted range of temperature increase per stage[1, p 94] of 33-50K in order to find the required number of stages. The calculated total temperature rise was of 295.3K, resulting in 7 stages being needed each with a temperature rise of 42.2K. The rest of the key results is shown in table 7 below, with the more detailed results

and dimensions in Appendix F.2 (total weight and length are 33kg and 0.313m). Note that due to the lower temperatures and thus greater allowable material stress, the structural limitations for the compressor proved to be far less constraining than those of the HPT, as expected.

Table 7: Key results of Compressor design

Inputs		Outputs	
RPM	28,100	α_2°	65.88
σ	1.1	Outlet Mach M_{3R}	0.322
M_1	0.55	Tip Mach M_r	0.818
α_1°	52.57	Axial Velocity V_{ax} (m/s)	186.9

7 Power Turbine Design

7.1 Design Process Overview

The methodology for the design of the PT is exactly the same as that for the HPT. However, while in the case of the HPT the RPM was limited only by material properties, in the case of the PT it is desirable to keep the RPM relatively low as the PT connects directly to the gearbox and thus the rotors, so a very high RPM would lead to a more complex and heavier gearbox[22]. In addition, although the temperature is lower than in the HPT, it is still too high for most materials to allow high rotational velocities, and thus to avoid having to use expensive materials like in the HPT a lower rotational velocity was preferred. The downside of a lower rotational velocity is that the value of Ω is lower, and as seen in figure 6 this means that the minimum achievable temperature ratio is higher. As a result the design decision came down to a compromise between having to use 2 stages and thus increasing weight, or using 1 stage but having to address the issues listed above.

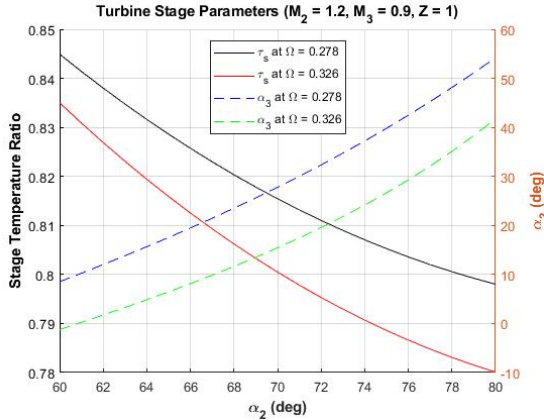
7.2 PT Design Results

First a single stage design was analysed, and two materials were considered, one being the same as used in the HPT, another being IN-939 equiaxed grain nickel-based alloy, which is lighter but has a lower yield strength[20] at these temperatures of 207 MPa, and would thus be limited at lower RPM. For a single stage design, assuming $M_2 = 1.2$ and $M_{3R} = 0.9$, a graph showing the attainable temperature ratio for each material as a function of α_2 is shown in figure 7 below.

As can be seen, the attainable Ω using the IN-939 alloy (0.278) has a minimum attainable temperature ratio of just under 0.8 (at the cost of very high α_3), which is insufficient for a single stage design, hence PWA 1484 must be used for a single stage. This can achieve the required temperature ratio at $\alpha_2 = 75.5^\circ$.

However the required RPM is 21,000 which is very high for a PT. Further, using the analysis presented in [21], it was found that this combination of parameters caused significant choking at the stator, hence the single stage option was discarded and a two stage design using the IN-939 alloy was selected instead. The key results are presented in table 8 below¹, with the full results and dimensions given in Appendix G (total PT weight and length 113.8kg and 0.13m).

¹The increase in ωr_{max} in stage 2 is due to the lower temperature leading to higher yield strength.

Figure 7: Attainable temperature ratios and α_3 values as a function of α_2 and Ω .**Table 8:** Key results of PT design

Parameters	Stage 1	Stage2
RPM	12,000	12,000
M_2	1.02	0.92
M_{3R}	0.8	0.8
α_2 °	60	60
α_3 °	18.81	29.2
Mean Radius r_m	0.23	0.24
ωr_{max} (m/s)	318	442
ωr actual (m/s)	270	278
DSF	1.44	0.79

8 Discussion, Conclusions & Overall Dimensions

Overall the engine design achieved and in some areas exceeded expectations, and grants the helicopter sufficient performance to successfully complete the mission. Due to a change in the rotor design at a late stage of the design [11], it cannot allow the helicopter to hover at 15,000ft with the payload as originally planned. Instead the ceiling with the payload is 14,000ft[12] (7000ft according to XPlane simulations carried out, however there were concerns with the accuracy of the modelling of the rotor[23]), however this was not deemed to be a significant issue as it can easily exceed this altitude without the payload (up to 27,000ft[23]), and thus does not affect the helicopter's ability to perform the capture maneuver.

In the end the choice of a higher than usual burner outlet temperature resulted in an efficient and compact engine design, with a max diameter of 0.52m and a length² of 1.04m which is smaller than comparable engines such as the Pratt & Whitney PT6, General Electric T700, and Turbomeca Arriel[24], due to the lower mass flow rate required as the high temperature grants the engine high specific power. The overall engine weight is 205kg, which is slightly below the initial estimate from regression data of 210kg, showing that despite the decision of using a two stage PT the weight penalty was not too significant, although it would have been approximately 20kg lighter. The resulting power to weight ratio of 6kW/kg is still however a very competitive value for turboshafts. Several additional graphs showing various parameters of engine performance under different operating conditions were also generated using GasTurb13, in Appendix H, including the power lapse, which exceeded expectations from the initial prediction obtained using the method in Prouty's[10] (see Appendix B).

The high burner temperature however required the use of advanced materials in the turbine section, as well as blade cooling, but the performance gains make it a competitive design that can be used on other future designs or retrofitted on existing rotorcraft. For a second design iteration, the possibility of pushing this concept further by raising the burner temperature to 1800-1900K should be considered. This would increase overall performance, however would require the use of materials such as Ceramic Matrix Composites (CMCs) for the turbine blades. This comes with significantly increased manufacture cost, but the resulting increases in power to weight ratio (due to the CMC's much lower weight) and SFC would give the engine a very good competitive edge.

²See relevant full component results in Appendices for breakdown of length and weight

Table 9: *Summary of Key Vehicle Specifications*

Parameter	Value (Metric)	Value (Imperial)
Global Mass & Power		
Weight with Payload (MTOW)	2907 kg	6408 lb
Empty Weight, Helicopter	1500 kg	3307 lb
Fuel Capacity	500 kg	1102 lb
Payload Mass	1400kg	3086 lb
Power Required (Ceiling, ISA)	1193.12 kW	1600 hp
Rotor (For one Rotor)		
Number of Blades	2	2
Blade Mass (each per blade)	40 kg	88 lb
Rotor Diameter	8.05 m	26.4 ft
Rotor Aspect Ratio (0pt 0pt .32em 0pt.44em)	14	14
Rotor Solidity	0.0909	0.0909
Rotor Disk Loading	45.9 kg/m ²	9.4 lb/ft ²
Rotor Blade Loading	503 kg/m ²	103 lb/ft ²
Rotor Hover Tip Speed	205.4 m/s	673.9 ft/s (≈ 0.65 Mach at 20000ft)
Rotor Cruise Tip Speed	268.529 m/s	881 ft/s
Rotational Speeds		
Engine Output Shaft	1256.6 rad/s	12000 RPM
Rotor Speed	51.1 rad/s	487.5 RPM
Travel Speeds		
Cruise Speed	185 km/h, 51 m/s	99 kts
Cruise Speed (Loaded)	139 km/h, 38 m/s	73 kts
Max Speed (Tip Mach = 0.85)	225 km/h, 62 m/s	121 kts
Never Exceed Speed, v_{ne} (Tip Mach = 0.92)	243 km/h, 67 m/s	131 kts
Speed (Supersonic Flow, Tip Mach = 1)	264 km/h, 73 m/s	143 kts
Capture Velocity	144 km/h, 40 m/s	77 kts
Timings		
Total Mission Time	1 hr 49 mins	1 hr 49 mins

References

- [1] Stephen A. Suhr. *Preliminary Turbohaft Engine Design Methodology For Rotorcraft Applications*. Georgia Institute of Technology, 2006.
- [2] Jack D. Mattingly, William H. Heiser, and David T. Pratt. *Aircraft Engine Design*. American Institute of Aeronautics and Astronautics, 2002.
- [3] W. Tan Jo. Weight sizing, subsystems & hums for a novel synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Systems & Propulsion.
- [4] Z. Janusz. Weight sizing and subsystem design of a novel synchropter concept for the aerial recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Systems & Propulsion.
- [5] A. Sharma. Fuselage structural design of a novel synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Structures & CAD.
- [6] T. Malhotra. Generative structural analysis & cad of a novel synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Structures & CAD.
- [7] A. Bullitta. Costing, acoustic study and vehicle management overview of a remote piloted synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Coordination.
- [8] C. Gliner. Business case and vehicle management overview of a novel synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Coordination.
- [9] Y. Ghenima. Regulations and vehicle management overview of a remote piloted synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Coordination.
- [10] Raymond W. Prouty. *Helicopter Performance, Stability, and Control*. Krieger Publishing Company, 1986.
- [11] C. Wing Cheng. Rotor design of a novel synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Aerodynamics.
- [12] U. Tin Hang. Rotor blade modelling and optimisation for a novel synchropter concept for aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Aerodynamics.
- [13] Seyhan Gul, Ravi Lumba, Abhishek Shastry, Mrinalgouda Patil, Shashank Maurya, Amy Morin, Nishant Nemani, Anubhav Dr. Datta, Vengalattore Dr. Nagara, and Inderjit Dr. Chopra. *Extreme Altitude Mountain Rescue Vehicle*. Alfred Gessow Rotorcraft Center, Department of Aerospace Engineering, University of Maryland, 2019.
- [14] Son Luong. Lithium-ion battery. <https://www.cei.washington.edu/education/science-of-solar/battery-technology/>. Accessed: 13.06.2020.

-
- [15] C. E. Thomas. *Fuel Cell and Battery Electric Vehicles Compared*. H2Gen Innovations, Inc., 2009.
- [16] Anubhav Datta and Wayne Johnson. *Requirements for a Hydrogen Powered All-Electric Manned Helicopter*. NASA Ames Research Center, 2012.
- [17] John Schenderlein and Tyler Clayton. *Comparison of Helicopter Turboshift Engines*. University of Colorado, 2015.
- [18] [Author Unknown]. Jet engine design: Turbine cooling. <https://aerospaceengineeringblog.com/turbine-cooling/>, 2013. Accessed: 13.06.2020.
- [19] K. San Hon. Flight performance analysis of the novel synchropter - Individual Report. 2020. WRR Group Design Project, Flight Mechanics, Performance & Control.
- [20] [Author Unknown]. High-temperature, high-strength nickel base alloys. <https://www.stainless-steel-world.net/pdf/393.pdf>, 1995. Accessed: 13.06.2020.
- [21] Q H Prof. Nagpurwala. Design of axial flow turbine design of axial flow turbine-1. <http://164.100.133.129:81/econtent/Uploads/08-PT12-AxTurb-1%2053%20%5BCompatibility%20Mode%5D.pdf>, 2008. Accessed: 13.06.2020.
- [22] T. Cross. Transmission design for a novel uav synchropter for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Systems & Propulsion.
- [23] T. Wong. Simulation flight test (performance) - Individual Report. 2020. WRR Group Design Project, Simulations.
- [24] Paul Jackson. *Jane's All the World's Aircraft*. Jane's Information Group, 2009.
- [25] Daniel P. Raymer. *Aircraft Design, A Conceptual Approach*. merican Institute of Aeronautics and Astronautics, 2018.
- [26] T. Walker. Subsystem layout of a novel synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Systems & Propulsion.
- [27] A. R. Polvara. Fuselage aerodynamics and modelling of a novel synchropter concept for the aerial-recovery of falling rocket stages - Individual Report. 2020. WRR Group Design Project, Aerodynamics.
- [28] Brian Wang. Siemens and emrax claim best power to weight ratio for electric motors in the 5 to 10 kilowatt per kg range. <https://www.nextbigfuture.com/2015/04/siemens-and-emrax-claim-best-power-to.html>, 2015. Accessed: 13.06.2020.
- [29] Robert E. Jones, Arthur M. Trout, and Jerrold D. Wear. *Combustion Gas Properties of Various Fuel of Interest to Gas Turbine Engineers*. Lewis Research Center, NASA, 1984.
- [30] Richard L. Turner and Albert M. Lord. *Thermodynamic Charts For The Computation Of Combustion And Mixture Temperatures At Constant Pressure*. National Advisory Committee for Aeronautics, 1946.
- [31] Svetlana Senkova, Oleg Senkov, and Dan Miracle. Cryogenic and elevated temperature strengths of an al-zn-mg-cu alloy modified with sc and zr. *Metallurgical and Materials Transactions A*, 37:3569–3575, 12 2006.
-

- [32] [Author Unknown]. Grade 6al 6v 2sn titanium alloy (uns 56620). <https://www.azom.com/article.aspx?ArticleID=9306>, 2013. Accessed: 13.06.2020.

A Pre-Conceptual Design

A.1 Pre-Conceptual Design Introduction

In this section the work done during the pre-conceptual phase is described. The WRR group was split into 3 sub-groups, each of which was to study one of the three options considered: A fixed wing aircraft, a VTOL aircraft, and a rotorcraft. This section will focus on the aircraft sub-group.

A.2 Baseline Configuration

Although several different recovery mechanisms were considered by the group, all but one revolved around the use of a military transport style aircraft. As a result, the baseline throughout the calculations is assumed to be a small military transport aircraft of conventional configuration, similar to an EADS/CASA C-295[24]. In addition, due to the relatively low speed required for the capture maneuver (stated as 45m/s at this stage), jet propulsion was deemed unnecessary, especially due to the relatively small size of the aircraft and thus lower power requirements. The remaining options were a turboprop, piston engines, or an electric design, which were compared in a similar fashion to what was later done for the rotorcraft.

A long range was required in order to fly to the recovery location (several hundred km offshore) and then return to base, as well as a ceiling of 30,000ft in order to have more time to capture the payload as it descends. The range requirement, in addition to the weight of the payload that needed to be carried made the choice of an electric design unsuitable due to the very low relative energy density of batteries, hence in the end the turboprop option was selected due to increased power to weight ratio and reliability.

A.3 Initial Sizing Calculations

After the baseline configuration had been determined to a sufficient degree work on initial sizing began. The calculations were carried out following the methods set out in Raymer's Aircraft Design [25].

Engine SFC data was taken from the Rolls-Royce AE2100 Turboprop engine, and several aerodynamic values such as aspect ratio and $\frac{S_{wet}}{S_{ref}}$ were taken from the CN-235 as a first approximation. The remaining parameters required were then calculated, with wing loading determined by the required stall speed in order to be able to safely fly at the recovery speed with a 20% safety margin. The resulting MTOW for a range of 2250km and 30 minutes of loitering was 9 tonnes.

B Power Estimation Calculations

The methods used to estimate power requirements throughout the various phases of flight are explained here. These were used for engine power sizing as well as fuel estimation for the final MTOW calculation [3], which required an estimate of power consumption. In addition, several forms of power losses were accounted for as necessary, following the methods from [10, p 275-278], taking the lower bound estimates for a more conservative result. Losses due to overboard bleed air were not considered as no components utilise this on board. For losses due to electrical generators, a maximum of 50kW of electrical power were considered [26]. Power lapse due to altitude and temperature differences were accounted for using the following equation from [10].

$$HP_{available} = HP_{MaxASL}(1 - 0.195h_1)(1 - 0.005\Delta T_s) \quad (8)$$

Where h_1 is altitude in ft divided by 10,000, and ΔT_s is the difference in temperature from standard atmosphere in degrees Rankine.

B.1 Hovering Flight

The hovering flight calculation is the simplest and is used as a basis for the power calculations in other flight phases. The derivation, explained in [10], uses a momentum analysis of the airflow through the rotor, as in actuator disc theory. The power estimate is obtained as follows:

First, the required thrust must be calculated, using

$$T = \left(1 + \frac{D_v}{W}\right) W \quad (9)$$

Where D_v in lbf is the vertical drag caused by the rotor wake interaction with the fuselage (this was obtained from the aerodynamics team, as explained in [27], and W is the weight in lbs. The induced power required, in HP, was then calculated using:

$$P_i = \frac{T \sqrt{\frac{D.L.}{2\rho}}}{550} \quad (10)$$

Where $D.L.$ is the disc loading in lbf/ft², ρ is the density of the air in slug/ft³, and 550 is a conversion factor to horsepower. The induced power value is then divided by the figure of merit, which is a measure of the rotor efficiency (this was obtained again from the aerodynamics team [11] [12]), in order to get the actual required power.

B.2 Vertical Climb

The power required in a vertical climb depends on the area of the fuselage in and out of the rotor wake. In this case the fuselage is entirely in the rotor wake, which allowed simplification of the calculation, with power required being given by:

$$P = W(v_{1c} + V_C) + 4 \left(\frac{D_{v_{hov}}}{W} \right) \frac{\rho}{2} (v_{1c} + V_C)^3 A_M \quad (11)$$

Where V_C is the vertical climb speed in ft/s, A_m is the main rotor area in ft², and v_{1c} is the induced downwash velocity in ft/s, calculated as follows:

$$v_{1c} = -\frac{V_C}{2} + \sqrt{\left(\frac{V_C}{2}\right)^2 + \frac{T}{2\rho A_M}} \quad (12)$$

Where T is the thrust in hover in lbf.

B.3 Forwards flight

The induced power required for forwards flight can be calculated as

$$P_i = \frac{T^2}{1,100\rho A_M V e} \quad (13)$$

Where V is forwards flight speed in knots and e is an efficiency factor, equivalent to Oswald efficiency for fixed wing aircraft. The thrust T in lbf is obtained from:

$$T = 2v_1\rho A_M \sqrt{V^2 + v_1^2} \quad (14)$$

Where the rotor induced velocity, v_1 in ft/s is obtained as follows:

$$v_1 = \sqrt{-\frac{V^2}{2} + \sqrt{\frac{V^4}{4} + \left(\frac{T_{hov}}{2\rho A_M}\right)^2}} \quad (15)$$

To find the actual power, the parasite drag needs to be accounted for, as follows:

$$P_p = \frac{D_p V}{550} \quad (16)$$

The parasite power P_p is then added to the induced power to obtain the total.

C Conceptual Engine Design

C.1 Engine Type Comparison

Table 10: Comparison of rotorcraft powerplant types

Powerplant type	Advantages	Disadvantages
Turboshaft	High power to weight ratio. Widely used and proven technology. Reliable	More expensive. Higher fuel consumption. Power lapse with altitude.
Piston	Less costly. Widely used and proven technology. Relatively low fuel consumption.	Less reliable. Lower power to weight ratio. Increased power lapse with altitude.
Electric	Much simpler engine/transmission. No power lapse. No emissions.	Novel technology in aviation, so far only applied to small aircraft. Batteries have very low energy density, leading to overall lower payload fractions, and lower range.

C.2 Electric Engine Weight Estimate

In order to calculate an estimate of the weight for an electric propulsion system, the following data were used:

- Electric motor power/weight ratio: 10kW/kg[28]
- Maximum power required 1450HP (1080kW)
- Battery energy density: 1MJ/kg[14]
- Average power consumption: 360kW
- Mission duration: 2h

To obtain the estimated weight of the motor, the maximum power required is divided by the power to weight ratio, obtaining a weight of 108kg. To estimate the weight of the batteries, the total energy requirement is estimated as $AveragePower \times MissionDuration(s)$, yielding a requirement of 2.59GJ. Dividing this by the energy density, a weight of about 2600kg is obtained.

C.3 Comparison of Piston and Turboshift Engines

Table 11: Power to weight ratios of helicopter piston (P) and turbine (T) engines.

Engine Type (Turbine/Piston)	Power to weight ratio (kW/kg)
Lycoming O-435 (P)	0.80
Lycoming O-360 (P)	1.15
Lycoming O-540 (P)	1.12
Zanzottera MZ 201 (P)	1.24
Pratt & Whitney (P&W) T73 (T)	7.66
Lycoming T53 (T)	3.47
P&W Canada PT6 (T)	3.52
Turbomeca Arriel (T)	5.00

For the calculation comparing the diesel and turboshift engines, the following data were used:

- Diesel engine power to weight ratio: 1.2kW/kg (best value from table 11)
- Turboshift power to weight ratio 3.5kW/kg (lowest value from table 11)
- Diesel SFC: 0.22kg/kWh (Austro Engine E4 [24])
- Turboshift SFC: 0.34kg/kWh ([17])
- Maximum power requirement diesel (no power lapse): 1450HP (1080kW)
- Maximum power requirement turboshift: 2400HP (1790kW)
- Average power requirement diesel: 360kW
- Average power requirement turboshift: 600kW
- Mission duration: 2h

First the weight of each engine was calculated by dividing the maximum power requirement by the power to weight ratio. Then the fuel mass required was estimated by $SFC \times MissionDuration \times AveragePower$ for each engine.

C.4 Hybrid System Calculation

The calculations follow the same methods and assumption used for the electric system and the turboshift calculation above, with the exception of the max power requirement being 1600HP (1194kW), and the average thus being around 400kW. The weight of the turboshift engine is obtained from regression data as:

$$W = 6.8916(HP_{max})^{0.5684} \quad (17)$$

Where the weight obtained is in lbs. Configurations with electric motors in intervals of 100HP from 100 to 500HP were considered. The power of the electric motor is subtracted from the max power required of the turboshift, and the expected maximum operating time of the electric motor was 30 minutes, which is the upper limit for the additional power requirements in helicopter engines[1, p 52].

C.5 Turboshaft Configurations

C.5.1 Single Spool

The single spool design is the simplest, with the compressor and turbine on the same spool thus with the same rotational velocity, and the turbine powering the compressor as well as providing the output power. The advantages of this design are the lower complexity and costs, however the performance is limited due to the output RPM being constrained to relatively low values due to needing to interface with the gearbox, which thus limits the efficiency and performance of the compressor and turbine.

C.5.2 Dual Spool

The dual-spool (also known as single spool with free turbine) configuration is essentially the same as the single spool however the turbine has been split into a high pressure turbine (HPT), which is on the compressor spool and powers the compressor only, and a power turbine (PT) which provides the output shaft power and is on a separate spool. This has the advantage of allowing the PT to rotate independently, allowing more flexibility and a better optimised design.

C.5.3 Three Spool

The three spool design adds a secondary spool to the both the compressor and gas turbine, in addition to retaining the free power turbine. This configuration is more mechanically complex and thus more costly and heavy, however the advantage is that splitting the compressor and gas turbine spools into low and high pressure spools allows a further optimised compressor design, thus being able to achieve higher pressure ratios without risks of surge.

D Initial Cycle Design

D.1 Initial Cycle Design Script

The following script was used for the initial cycle analysis. Values of γ and C_p at different temperatures and fuel-air ratios were obtained from [29].

```

1
2 % AERO96005 GROUP DESIGN PROJECT
3 % Engine Sizing
4 % -----
5 % DETERMINES ENGINE MASS FLOW, FUEL-AIR RATIO, AND TOTAL TEMPERATURES/PRESSURES ...
   FOR GIVEN INLET CONDITIONS, ADIABATIC EFFICIENCIES, COMPRESSOR PRESSURE ...
   RATIO, AND BURNER EXIT TEMPERATURE.
6 % OMAR ARANGATH 20/05/2020
7 % -----
8
9 clear
10 clc
11
12 gamma = 1.4;
13 %% INLET
14 Ta = 300;%Ambient temperature
15
16 Pa = 101325;%Ambient pressure

```

```

17
18 InletPresRatio = 0.97;
19
20 Pod = Pa * InletPresRatio;
21
22 Tod = Ta * InletPresRatio^(gamma/(gamma-1));
23
24 Δ.diff = Pod/Pa;
25
26 theta.diff = Tod/Ta;
27 %% COMPRESSOR
28 Cp = 1004.7;
29
30 CompressorPresRatio = 10;
31
32 CompressorAdiabaticEfficiency = 0.85;
33
34 CompressorTempRatio = ((CompressorPresRatio^((gamma-1)/gamma) - ...
    1)/CompressorAdiabaticEfficiency) + 1;
35
36 CompressorSpecificWork = Cp * Ta * (CompressorTempRatio - 1);
37
38 Poc = Pod * CompressorPresRatio;
39
40 Toc = Tod * CompressorTempRatio;
41
42 Δ.comp = Poc/Pa;
43
44 theta.comp = Toc/Ta;
45 %% BURNER
46 BurnerPressureRatio = 0.92;
47
48 Pob = Poc * BurnerPressureRatio;
49
50 Tob = 1700;%Burner exit temperature
51
52 FuelEnergy = 44.5e6;
53
54 Cp2 = 1156.2;%1700K
55 gamma2 = 1.275;%1700K
56
57 airfuelratio = (FuelEnergy/(Cp2*Tob - Cp*Toc)) - 1;
58
59 fuelairratio = 1/airfuelratio;
60
61 burnerefficiency = ((airfuelratio + 1) * Cp2 * Tob - airfuelratio * Cp * ...
    Toc)/FuelEnergy;
62
63 Δ.burn = Pob/Pa;
64
65 theta.burn = Tob/Ta;
66
67 %% HP Turbine
68 HPTurbineAdiabaticEfficiency = 0.85;
69
70 Tot1 = Tob - (CompressorSpecificWork/((1 + fuelairratio) * Cp2));
71
72 HPTurbineTempRatio = Tot1/Tob;
73

```

```

74 HPTurbinePressureRatio = (1 - ...
    (1-HPTurbineTempRatio)/HPTurbineAdiabaticEfficiency)^(gamma2/(gamma2-1));
75
76 Pot1 = Pob * HPTurbinePressureRatio;
77
78 Δ.turb1 = Pot1/Pa;
79
80 theta.turb1 = Tot1/Ta;
81 %% LP Turbine/Exhaust
82
83 gamma3 = 1.29;%1700K Tob
84
85 ExhaustPressureRatio = 0.98;
86
87 Pot2 = Pa/ExhaustPressureRatio;
88
89 LPTurbinePressureRatio = Pot2/Pot1;
90
91 LPTurbineAdiabaticEfficiency = 0.85;
92
93 LPTurbineTemperatureRatio = ((LPTurbinePressureRatio^((gamma3-1)/gamma3) - 1) * ...
    LPTurbineAdiabaticEfficiency) + 1;
94
95 Tot2 = Tot1 * LPTurbineTemperatureRatio;
96
97 ShaftSpecificWork = (1 + fuelairratio) * Cp2 * (Tot1 - Tot2);
98
99 requiredshaftpower = 1195e3 + 50e3;%Required power plus electrical generation.
100
101 requiredmassflow = requiredshaftpower/ShaftSpecificWork;
102
103 Δ.turb2 = Pot2/Pa;
104
105 theta.turb2 = Tot2/Ta;
106
107 mfuel = requiredmassflow * fuelairratio;
108
109 MassFlow = requiredmassflow - mfuel;
110
111 SFC = mfuel * 3600/requiredshaftpower;

```

D.2 Turbine Cooling Details

The cooling efficiency factor E needs to be adjusted by the pattern factor, PF which accounts for differences in flow direction depending on the position in the engine[1, p 168]. This is done as follows:

$$E_{PF} = \frac{PF + E}{PF + 1} \quad (18)$$

Where PF is the pattern factor, taken as 0.3 for the first row of vanes and 0.13 everywhere else. In order to obtain the flow temperature after the first introduction of cooling flow, an enthalpy calculation is carried out as follows.

$$h_{4,1} = \frac{h_4 \dot{m}_4 + h_c \dot{m}_c}{\dot{m}_{4,1}} \quad (19)$$

Where subscript 4,1 refers to post-cooling condition, subscript 4 refers to pre-cooling conditions, and subscript c refers to the cooling air flow. Note that $\dot{m}_{4,1} = \dot{m}_4 + \dot{m}_c$. Following this, using the known fuel mass flow rate the local fuel to air ratio can be calculated, and using this data and enthalpy tables from [30], the temperature at the rotors can be estimated. This is then used as for the stator calculation to obtain the required cooling mass flow rate.

E Turbine Design & Analysis

E.1 Turbine Graphs

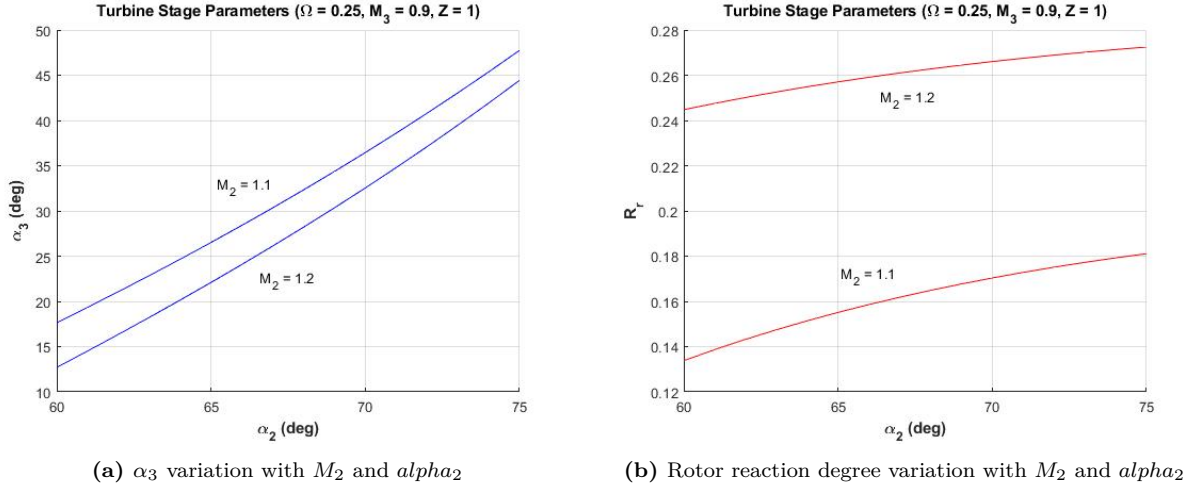


Figure 8: Effects of varying inlet mach number relative to blades M_2 on outlet swirl angle α_3 and rotor reaction degree R_r .

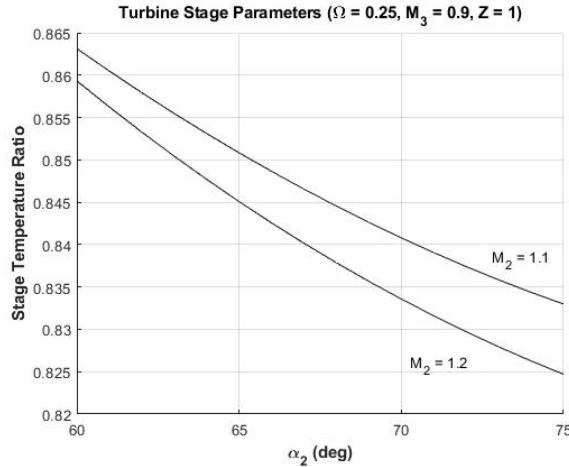


Figure 9: Effects of varying inlet mach number relative to blades M_2 on stage temperature ratio

The flow turning angle $\Delta\beta$, which under accepted convention should remain under 130° assuming that the rotor degree of reaction is at least 0.2[2, p 277] was found not to vary significantly with M_2 . The parameters shown in the figure titles are non-dimensional rotational velocity Ω from equation

4, the exit relative mach number M_3 , and Zweifel Coefficient Z which is a measure of how close to ideal the pressure distribution on the blade is in order to avoid separation. This is given by:

$$Z = \frac{2\cos^2\beta_3(\tan\beta_2 + \tan\beta_3)}{\sigma} \quad (20)$$

Where β_2 and β_3 are the rotor relative inlet and exit angles respectively, and σ is the rotor solidity.

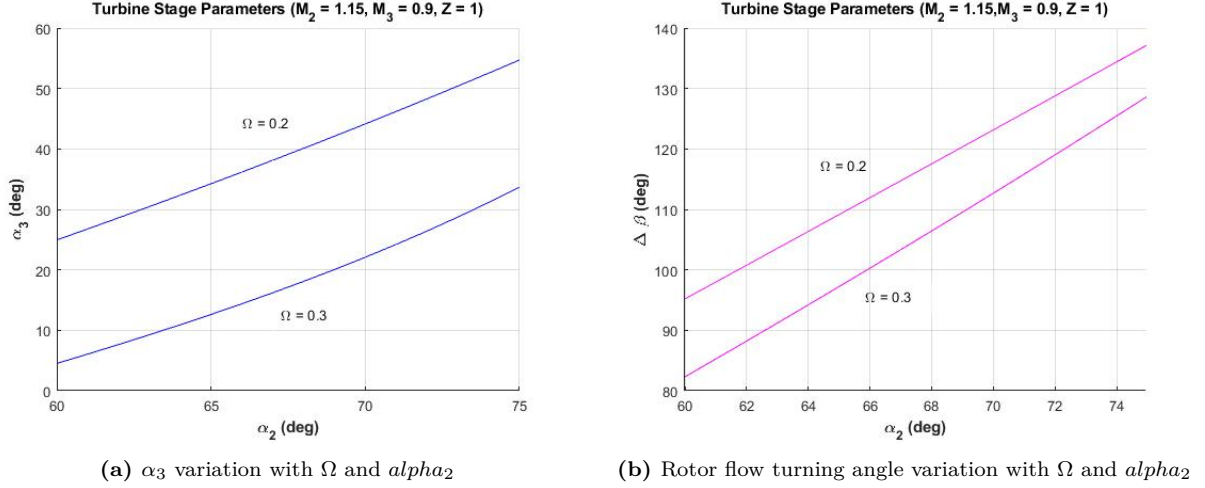


Figure 10: Effects of varying Ω on various turbine parameters

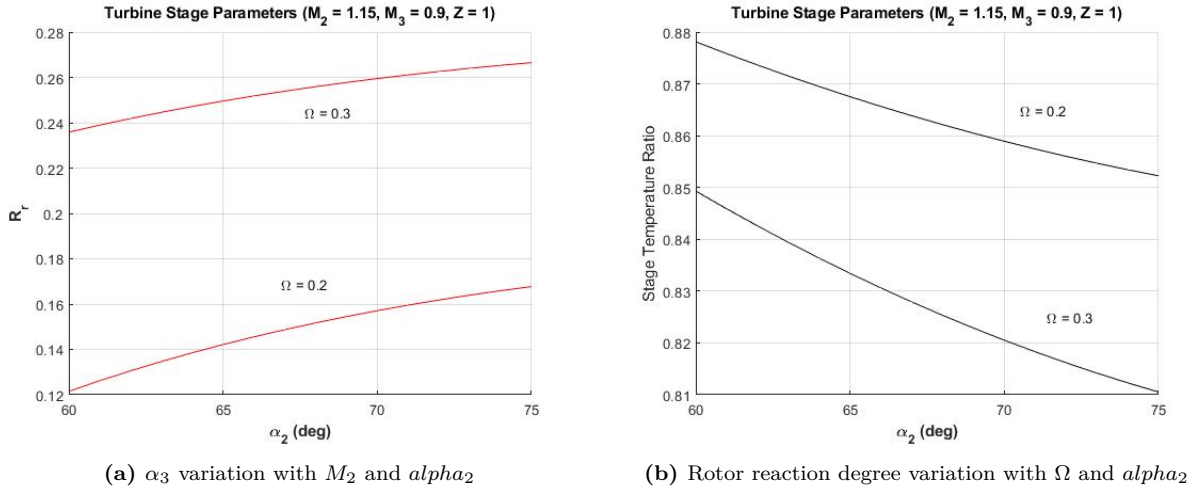


Figure 11: Effects of varying Ω on various turbine parameters

E.2 HPT Design Results & Dimensions

The final results obtained for the HPT are given here:

Table 12: *Final HPT Design Results*

Mass flow rate (kg/s)	3.383
Stage flow coefficient Φ	0.4813
Stage loading coefficient Ψ	2.422
Axial Velocity (m/s)	197.2
Outlet Mach M_3	0.3392
Inlet Annulus Area (m²)	0.0077
Stator Annulus Area (m²)	0.0116
Rotor Annulus Area (m²)	0.0169
AN_{max}^2	2.38e8
AN^2	1.43e8

Although Ψ falls within the range given in [21], the values of Φ and M_3 are slightly below standard values, due to the relatively high α_2 and α_3 , and although does not stop the turbine from performing its function, it is something to be improved in the next design iteration. The AN_{max}^2 factor is another limitation calculated as follows:

$$\Omega = \frac{3600\sigma_c}{\pi\rho\left(1 + \frac{A_t}{A_h}\right)} \quad (21)$$

Where $\frac{A_t}{A_h}$ is the taper ratio, assumed to be 0.8[2, p 292], σ_c is material stress in ksi, and ρ is the material density in slug/ft³ (17.35 in this case). The actual value of AN^2 is then calculated by simply multiplying the flowpath area at the rotor by the RPM².

Turbine dimensions are given here:

Table 13: *Final HPT Design Dimensions*

Parameter	Stator	Rotor
Annulus (blade) height (m)	0.0132	0.0193
Tip radius r_t	N/A	0.149
Disc + rim radius r_h	N/A	0.130
Number of vanes/blades	146	94
Blade Chord (m)	0.0264	0.0116
Chord/Height ratio	2	0.6
Reaction Degree	0.765	0.235
Turbine rim height h_r (m)	N/A	0.0064
Disc radius r_r (m)	N/A	0.123
Rim flange to Disc thickness ratio W_{dr}/W_r	N/A	0.47
Rim thickness W_{dr} (m)	N/A	0.0065
Disc base thickness W_{ds} (m)	N/A	0.0805
Rim flange thickness	N/A	0.0127

All these values fall within the ranges specified in [1, p 183-190], indicating a reasonable design. The weight and length of the turbine were estimated as in [1, p 189], that is by taking the turbine as a thin disk with the thickness varying from W_{ds} to W_{dr} . The results for the HPT were a weight of 48.6kg and a length of 0.087m. See diagram below for for diagram indicating dimensions.

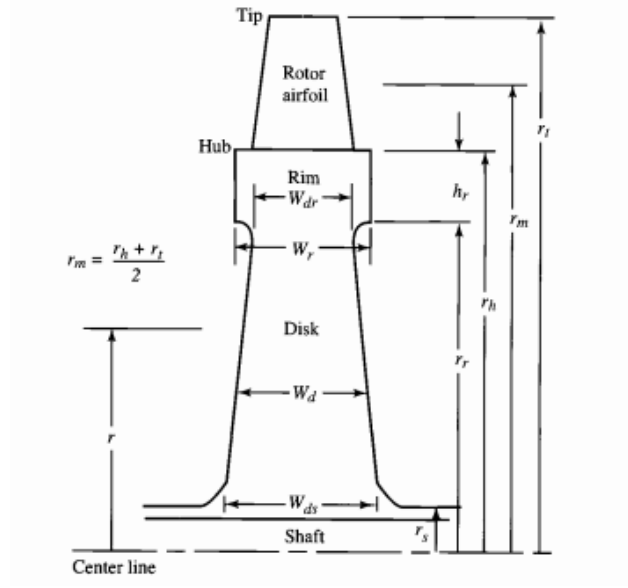
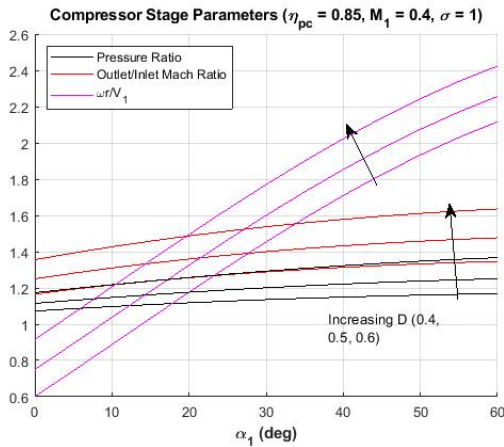


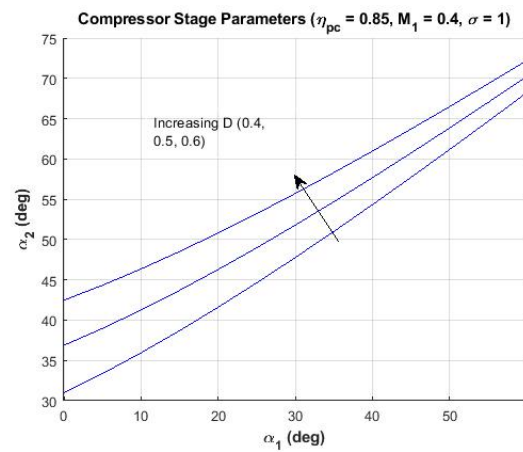
Figure 12: Diagram explaining different turbine dimensions.[2, p 290]

F Compressor Design & Analysis

F.1 Compressor Graphs



(a) Pressure ratio and inlet outlet Mach ratio variation with D and α_{α_1}



(b) α_{α_2} variation with D and α_{α_1}

Figure 13: Effects of varying D on various compressor parameters

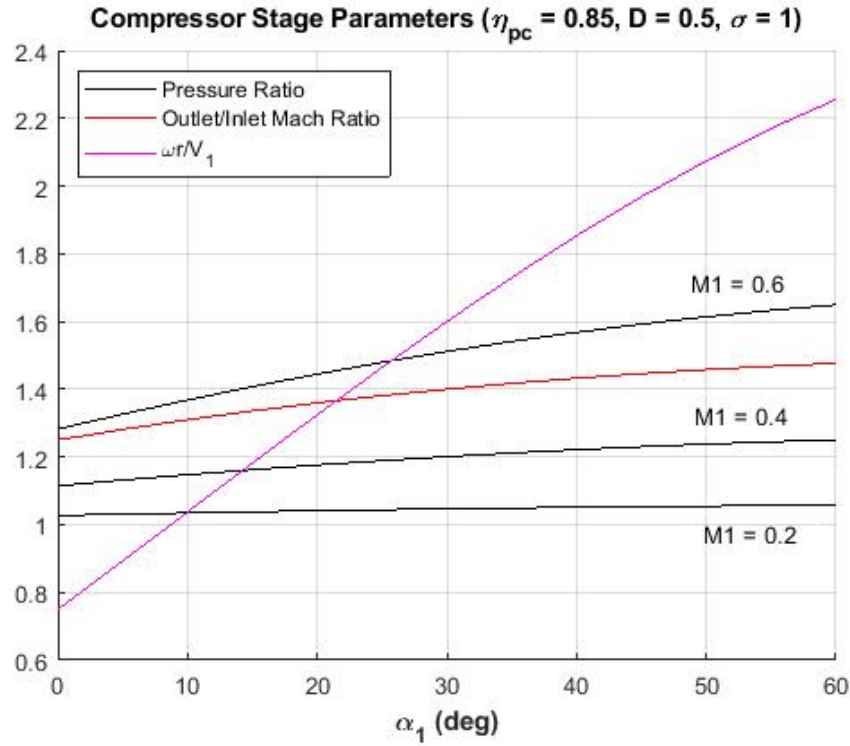
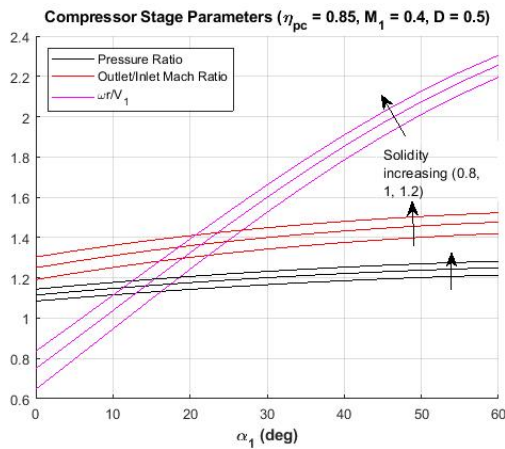
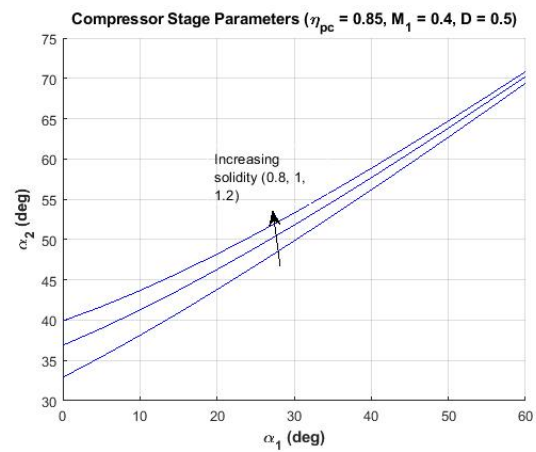


Figure 14: Effects of varying M_1 on pressure ratio



(a) Pressure ratio and inlet outlet Mach ratio variation with σ and α_{11}



(b) α_2 variation with σ and α_{11}

Figure 15: Effects of varying σ on various compressor parameters

F.2 Compressor Design Results & Dimensions

Table 14: *Final Compressor Design Parameters*

Chord/Height ratio of rotor blades	0.3
Chord/Height ratio of stator blades	0.8
Rotor Loss Coefficient Φ_{rot}	0.1
Stator Loss Coefficient Φ_{stat}	0.06
Shaft Radius (m)	0.02
$\Delta\alpha$ (deg)	13.31
Temperature Ratio τ_s (per stage)	1.156
Pressure Ratio π_s for 5 repeating stages	1.54
Pressure Ratio π_s for 2 non-repeating stages	1.0744
$\omega r/V_1$	2.152
M_r/M_1 (Tip Mach to Inlet Mach ratio)	1.487
M_3/M_1 (Outlet to inlet Mach ratio)	0.926
ω_{rmax} (m/s)	654.3
Highest ω_r (stage 7) (m/s)	386.2
Stage flow coefficient Φ	0.2825
Stage loading coefficient Ψ	0.2619
DSF (max, stage 7)	0.5775
Airfoil camber angle	16.49
Airfoil deviation	3.17

As can be seen, the structural limitations on the compressor are much less restrictive than in the HPT. The material used on the first 5 stages is a modified Aluminium 2618-T6 alloy, which is possible since the temperature at the 5th stage does not exceed 230 degrees Celsius, at which the yield strength of the alloy is 190MPa [31]. For the last two stages the higher temperature makes the use of a titanium alloy (6Al 6V 2Sn Titanium Alloy (UNS 56620)) necessary. This has a yield strength of 586 MPa [32] at the operating temperature.

Table 15: *Final Compressor Design Dimensions*

	Stage 1	Stage 7
Annulus area (m²)	0.0426	0.0093
Annulus (blade) height (m)	0.0496	0.0109
Tip radius r_t (m)	0.1614	0.1421
Disc + rim radius r_h	0.1119	0.1312
Number of vanes	24	109
Number of blades	39	174
Blade Chord (m)	0.0248	0.0054
Vane Chord (m)	0.0396	0.0087
Rim thickness W_{dr} (m)	0.0245	0.0054
Disc base thickness W_{ds} (m)	0.0339	0.0105

The compressor values shown are given for stages 1 and 7, as they show the variation of values across the compressor. Again the resulting values fall within the ranges stated in [1, p 183-190].

The estimated compressor weight and length are 33kg and 0.313m, with the length estimated using the discs as in the turbine in addition to the estimated axial spacing between stages, estimated as a quarter of the previous rotor/stator chord [2, p 264]. The chord to height ratio used was 0.5 for the rotor and 0.8 for the stator.

G Power Turbine Results & Dimensions

Table 16: *Final PT Design Results*

	Stage 1	Stage 2
Mass flow rate (kg/s)	3.545	3.545
Stage flow coefficient Φ	1.186	1.218
Stage loading coefficient Ψ	4.786	3.445
Axial Velocity (m/s)	325	285
Outlet Mach M_3	0.5429	0.6860
Inlet Annulus Area (m²)	0.0121	N/A
Stator Annulus Area (m²)	0.0187	0.230
Rotor Annulus Area (m²)	0.0215	0.309
AN_{max}^2	1.73e8	3.35e8
AN^2	3.33e7	4.79e7

Table 17: *Final PT Design Dimensions*

Parameter	Stator 1	Rotor 1	Stator 2	Rotor 2
Annulus (blade) height (m)	0.0129	0.0148	0.0151	0.0203
Tip radius r_t	N/A	0.238	N/A	0.252
Disc + rim radius r_h	N/A	0.223	N/A	0.232
Number of vanes/blades	232	176	222	140
Blade Chord (m)	0.0232	0.0148	0.0241	0.0162
Chord/Height ratio	1.8	1	1.6	0.8
Reaction Degree	0.207	0.793	0.819	0.181
Turbine rim height h_r (m)	N/A	0.0082	N/A	0.0089
Disc radius r_r (m)	N/A	0.215	N/A	0.224
Rim flange to Disc thickness ratio W_{dr}/W_r	N/A	0.434	N/A	0.401
Rim thickness W_{dr} (m)	N/A	0.0071	N/A	0.0073
Disc base thickness W_{ds} (m)	N/A	0.0686	N/A	0.0398
Rim flange thickness	N/A	0.0163	N/A	0.0178

Again values fall within the acceptable ranges, with the exception of the Ψ value on the first stage which is abnormally high, in addition to the exit Mach number being quite high, so these would be areas to address in the next iteration. The ideal re-design of the PT would use a more advanced material and possibly more cooling flow in order to achieve a single-stage design. The dimensions are:

- Stage 1 weight: 52.64kg
- Stage 1 length: 0.0781m

- Stage 2 weight: 61.2kg
- Stage 2 length: 0.0499m

The total weight of the turbomachinery components is thus 195.4kg and the total length (including the burner length of 0.244m [3]) is 0.789m. Assuming an additional length of 25cm for spacing between components, a full length of 1.04m is obtained, and adding 10kg to account for shafts, casing and the burner, the estimate of 205kg is obtained.

H GasTurb 13 Diagrams

H.1 Performance

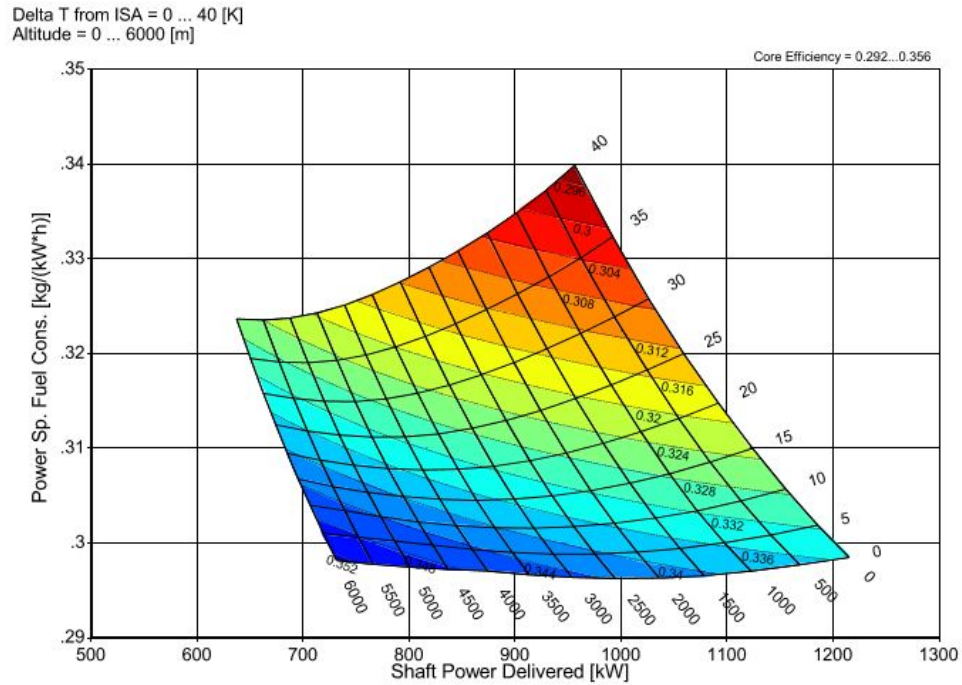
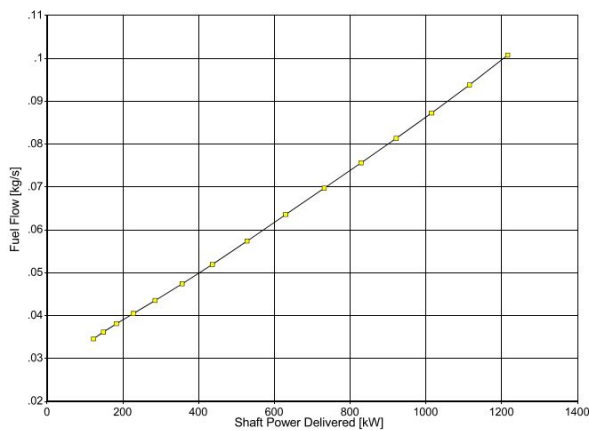
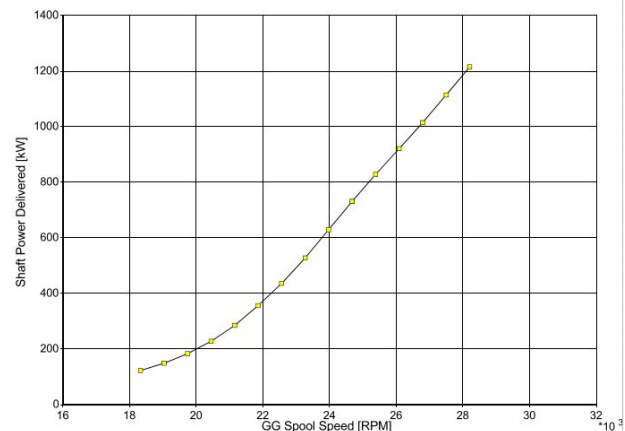


Figure 16: Engine performance (SFC & power available) as a function of altitude and temperature.

H.2 Other



(a) Variation of output power with fuel flow (at sea level).



(b) Variation of output power with high pressure spool RPM.

Figure 17: Power variation with RPM and fuel flow.

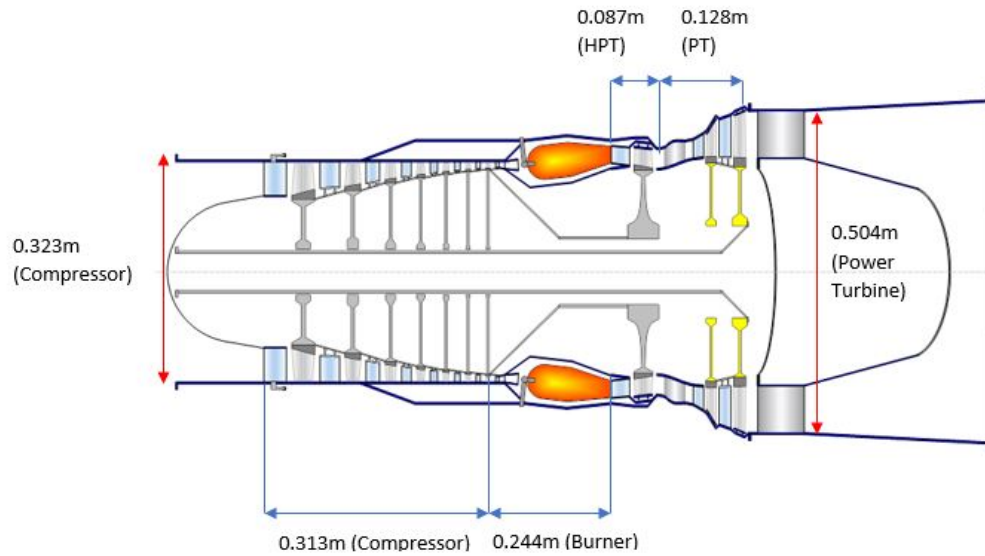


Figure 18: Key Engine Dimensions

H.3 Compressor and Turbine Maps

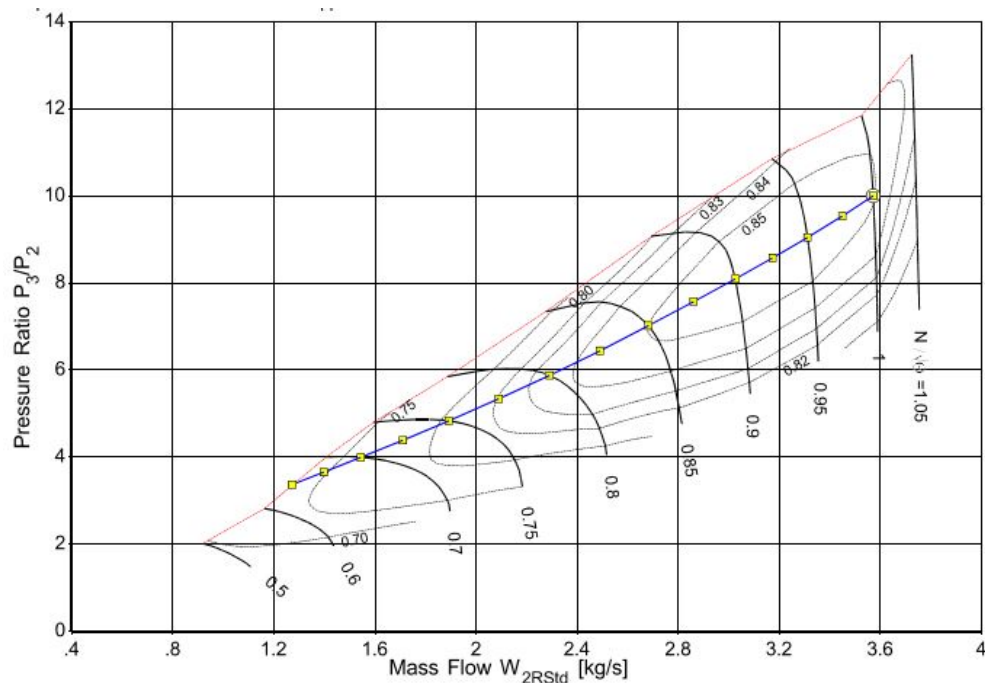


Figure 19: Compressor map

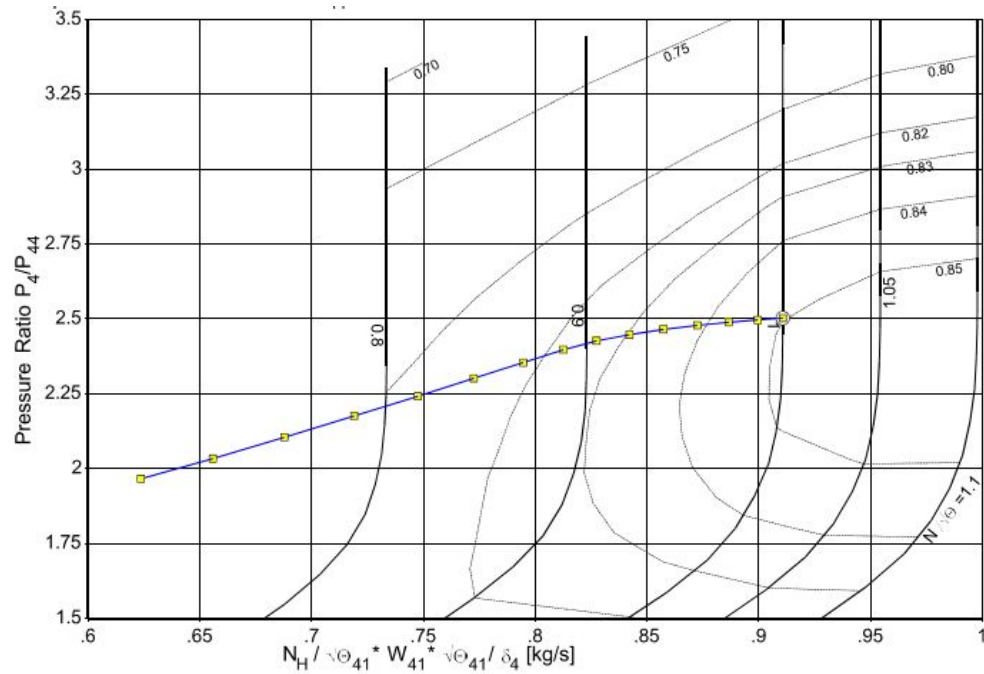


Figure 20: High Pressure Turbine Map

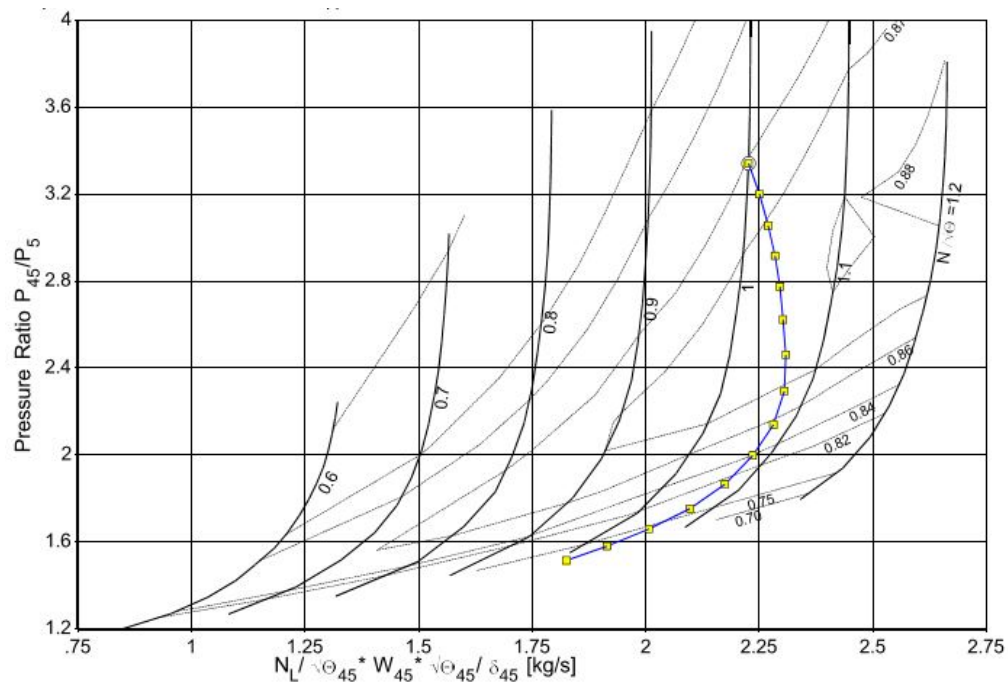


Figure 21: Power Turbine Map